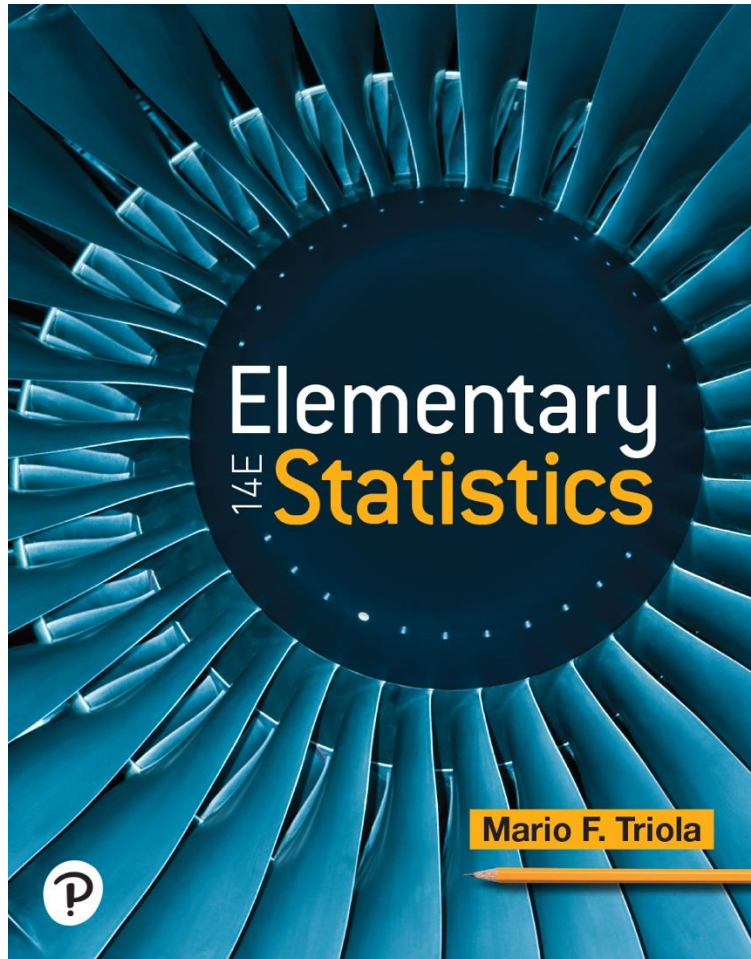


Elementary Statistics

Fourteenth Edition



Chapter 4 Probability

Probability

4-1 Basic Concepts of Probability

4-2 Addition Rule and Multiplication Rule

4-3 Complements and Conditional Probability

4-4 Counting

Key Concept (1 of 2)

In this section we present the **addition rule** as a tool for finding $P(A \text{ or } B)$, which is the probability that either event A occurs or event B occurs (or they both occur) as the single outcome of a procedure. The word “or” in the addition rule is associated with the addition of probabilities.

Key Concept (2 of 2)

This section also presents the basic **multiplication rule** used for finding $P(A \text{ and } B)$, which is the probability that event A occurs and event B occurs. The word “and” in the multiplication rule is associated with the multiplication of probabilities.

Compound Event

- Compound Event
 - A **compound event** is any event combining two or more simple events.

Addition Rule

Notation for Addition Rule

$P(A \text{ or } B) = P(\text{in a single trial, event } A \text{ occurs or event } B \text{ occurs or they both occur})$

Intuitive Addition Rule

To find $P(A \text{ or } B)$, add the number of ways event A can occur and the number of ways event B can occur, but **add in such a way that every outcome is counted only once**. $P(A \text{ or } B)$ is equal to that sum, divided by the total number of outcomes in the sample space.

Formal Addition Rule

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

where $P(A \text{ and } B)$ denotes the probability that A and B both occur at the same time as an outcome in a trial of a procedure.

Example: Addition Rule (1 of 2)

At a public university, 40% of the students are taking introduction to statistics, 20% of the students are taking introduction to economics, and 8% of the students are taking both. What is the probability that a randomly selected student is taking introduction to statistics or introduction to economics?

Example: Addition Rule (2 of 2)

Solution

Using the addition rule, we get

$$\begin{aligned} P(\text{statistics or economics}) &= P(\text{statistics}) + P(\text{economics}) - P(\text{statistics and economics}) \\ &= 0.4 + 0.2 - 0.08 = 0.52 \end{aligned}$$

The probability that a randomly selected student is taking introduction to statistics or introduction to economics is 0.52.

Disjoint Events and the Addition Rule

- Disjoint (or mutually exclusive)
 - Events A and B are **disjoint** (or **mutually exclusive**) if they cannot occur at the same time. (That is, disjoint events do not overlap.)
 - Example: odd numbers and even numbers

Summary

Here is a summary of the key points of the addition rule:

1. To find $P(A \text{ or } B)$, first associate the word **or** with addition.
2. To find the value of $P(A \text{ or } B)$, add the number of ways A can occur and the number of ways B can occur, but be careful to add without double counting.

Complementary Events and the Addition Rule

We use $\bar{A}$ to indicate that event A does not occur. Common sense dictates this principle: We are certain (with probability 1) that either an event A occurs *or* it does not occur, so it follows that $P(A \text{ or } \bar{A}) = 1$. Because events A and $\bar{A}$ must be disjoint, we can use the addition rule to express this principle as follows:

$$P(A \text{ or } \bar{A}) = P(A) + P(\bar{A}) = 1$$

Rule of Complementary Events

$$P(A) + P(\bar{A}) = 1$$

$$P(\bar{A}) = 1 - P(A)$$

$$P(A) = 1 - P(\bar{A})$$

Example: Smartphone Home (1 of 2)

Based on survey results from the Consumer Technology Association, the probability of randomly selecting a household in the United States and getting one with a smartphone is 0.87, so $P(\text{smartphone}) = 0.87$. If a household is randomly selected, find the probability of getting one that does *not* have a smartphone.

Example: Smartphone Home (2 of 2)

Solution

Using the rule of complementary events, we get

$$\begin{aligned} P(\text{household } \textit{not} \text{ having a smartphone}) \\ &= 1 - P(\text{household with a smartphone}) \\ &= 1 - 0.87 = 0.13 \end{aligned}$$

The probability of randomly selecting a household *not* having a smartphone is 0.13.

Multiplication Rule

Notation

$P(A \text{ and } B) = P(\text{event } A \text{ occurs in a first trial and event } B \text{ occurs in a second trial})$

$P(B | A)$ represents the probability of event B occurring after it is assumed that event A has already occurred.

Intuitive Multiplication Rule

To find the probability that event A occurs in one trial and event B occurs in another trial, multiply the probability of event A by the probability of event B , but **be sure that the probability of event B is found by assuming that event A has already occurred.**

Formal Multiplication Rule

$$P(A \text{ and } B) = P(A) \cdot P(B | A)$$

Independence and the Multiplication Rule

- Independent
 - Two events A and B are **independent** if the occurrence of one does not affect the **probability** of the occurrence of the other. (Several events are independent if the occurrence of any does not affect the probabilities of the occurrence of the others.) If A and B are not independent, they are said to be **dependent**.

Example: Drug Screening and the Basic Multiplication Rule (1 of 6)

50 test results from the subjects who use drugs are shown below:

Positive Test Results: 45

Negative Test Results: 5

Total: 50

Example: Drug Screening and the Basic Multiplication Rule (2 of 6)

- a. If 2 of these 50 subjects are randomly selected **with replacement**, find the probability the first selected person had a positive test result and the second selected person had a negative test result.
- b. Repeat part (a) by assuming that the two subjects are selected **without** replacement.

Example: Drug Screening and the Basic Multiplication Rule (3 of 6)

Solution

a. With Replacement: First selection (with 45 positive results among 50 total results):

$$P(\text{positive test result}) = \frac{45}{50}$$

Second selection (with 5 negative test results among the same 50 total results):

$$P(\text{negative test result}) = \frac{5}{50}$$

Example: Screening Drugs and the Basic Multiplication Rule (4 of 6)

Solution

We now apply the multiplication rule as follows:

$P(\text{1st selection is positive and 2nd is negative})$

$$= \frac{45}{50} \cdot \frac{5}{50} = 0.0900$$

Example: Screening Drugs and the Basic Multiplication Rule (5 of 6)

Solution

b. Without Replacement: Without replacement of the first subject, the calculations are the same as in part (a), except that the second probability must be adjusted to reflect the fact that the first selection was positive and is not available for the second selection. After the first positive result is selected, we have 49 test results remaining, and 5 of them are negative.

The second probability is therefore $5/49$.

Example: Screening Drugs and the Basic Multiplication Rule (6 of 6)

Solution

P (1st selection is positive and 2nd is negative)

$$= \frac{45}{50} \cdot \frac{5}{49} = 0.0918$$

Sampling

In the world of statistics, sampling methods are critically important, and the following relationships hold:

- Sampling **with replacement**: Selections are **independent** events.
- Sampling **without replacement**: Selections are **dependent** events.

Treating Dependent Events as Independent

5% Guideline for Cumbersome Calculations

When sampling without replacement and the sample size is no more than 5% of the size of the population, treat the selections as being **independent** (even though they are actually dependent).

Example: Drug Screening and the 5% Guideline for Cumbersome Calculations (1 of 3)

In a recent year, there were 130,639,273 full-time employees in the United States. If one of those employees is randomly selected and tested for illegal drug use, there is a 0.042 probability that the test will yield a positive result, indicating that the employee is using illegal drugs (based on data from Quest Diagnostics). Assume that three employees are randomly selected *without replacement* from the 130,639,273 employees in the United States. Find the probability that the three selected employees all test positive for drug use.

Example: Drug Screening and the 5% Guideline for Cumbersome Calculations (2 of 3)

Solution

Because the three employees are randomly selected without replacement, the three events are dependent, but here we can treat them as being independent by applying the 5% guideline for cumbersome calculations. The sample size of 3 is clearly no more than 5% of the population size of 130,639,273.

Example: Drug Screening and the 5% Guideline for Cumbersome Calculations (3 of 3)

Solution

We get $P(\text{all 3 employees test positive})$

$= P(\text{first test positive and second test positive and third test positive})$

$= P(\text{first test positive}) \cdot P(\text{second test positive}) \cdot P(\text{third test positive})$

$= (0.042)(0.042)(0.042) = 0.0000741$

There is a 0.0000741 probability that all three selected employees test positive.

Summary of Addition Rule and Multiplication Rule

- **Addition Rule for $P(A \text{ or } B)$:** The word **or** suggests addition, and when adding $P(A)$ and $P(B)$, we must add in such a way that every outcome is counted only once.
- **Multiplication Rule for $P(A \text{ and } B)$:** The word **and** for two trials suggests multiplication, and when multiplying $P(A)$ and $P(B)$, we must be sure that the probability of event B takes into account the previous occurrence of event A .